MODULE 06 · DOMAIN 5 - MODEL SELECTION AND OPTIMISATION

Shots, Thinking, Effort and Fast Mode

Showing Claude what you want with examples, giving it room to reason, and the dials that trade depth, speed and cost.

DOMAIN 5 of 8 EXAM WEIGHT 16.8% MODULE 6 of 36

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Two kinds of help: **show** Claude what good looks like, and give it room to **think**.

Zero, one or many shots

APPROACH	WHAT YOU GIVE	BEST FOR
Zero-shot	Instructions only	Clear, well-defined tasks
Single-shot	One worked example	Showing one exact format
Multi-shot	Three to five examples	Consistent format, tone, edge cases

THE TIEBREAKER

Start zero-shot. Add examples when the output varies in ways instructions cannot pin down.

What makes a good set of examples

- **Relevant:** mirror your real inputs closely.
- **Diverse:** cover edge cases, and vary enough that Claude does not copy unintended patterns.
- **Structured:** wrap each in **example** tags, inside an **examples** tag, apart from the instructions.
- **Have labelled examples already?** Put them in the prompt. No custom training needed.

> ON THE EXAM
>
> Anthropic: "A few well-crafted examples improve accuracy and consistency." Three to five is the sweet spot.

Multi-shot in practice: Copperlet's triage

```
<examples>
<example>
<ticket>Water dripping through kitchen ceiling from the flat above.</ticket>
<label>{"urgency": "urgent", "trade": "plumbing"}</label>
</example>
<example>
<ticket>Strong smell of gas in the hallway. We have left the flat.</ticket>
<label>{"urgency": "emergency", "trade": "heating"}</label>
</example>
<example>
<ticket>Kitchen cupboard door hinge is loose.</ticket>
<label>{"urgency": "routine", "trade": "other"}</label>
</example>
</examples>
```

7-8 An edge case: a gas smell is always an emergency.

11-12 A different urgency and trade, so no single pattern dominates.

THE TAKEAWAY

Same output shape every time, very different inputs. That is what diverse means.

Examples or instructions?

REACH FOR EXAMPLES WHEN

Show, don't tell

- Format or tone is hard to describe
- Known edge cases keep failing
- Labelled examples already exist
- Outputs vary in shape

REACH FOR INSTRUCTIONS WHEN

Tell, with a reason

- A rule has a clear reason
- There are many categories to define
- The logic must be explicit
- Examples would date quickly

THE TIEBREAKER

Use both: instructions say what and why, examples show how. If they disagree, Claude follows the examples.

When examples backfire

- **Copied too closely:** every output now has the examples' length, opening words or structure.
- **One-sided set:** all examples are routine, so emergencies get under-rated.
- **Contradiction:** examples show a format the instructions forbid.
- **Stale:** the output format changed, the examples did not.
- **Fix:** vary them, balance them, keep them consistent, and review them with the prompt.

> ON THE EXAM
>
> Examples are powerful because Claude imitates them. That is also their risk.

Thinking is **rough work** Claude does before answering. It usually helps on hard problems, and you pay for every token of it.

Thinking on today's models

MODEL	THINKING
Opus 5.5	Always on; cannot be turned off
Sonnet 5.5	On by default; disabling it is not allowed
Haiku 5.5	On by default; can be disabled at effort high or below
Opus 4.6 to 4.8, Sonnet 4.6	Off until you request adaptive thinking
Opus 4.5, Sonnet 4.5, Haiku 4.5	Extended thinking only, with a token budget

CURRENCY CHECK

On the newest models thinking is simply there. Your lever is effort, not an on-off switch.

Extended thinking or adaptive thinking?

EXTENDED THINKING

You set a budget

- type enabled plus budget_tokens
- Minimum budget 1,024 tokens
- The 4.5 models
- budget_tokens rejected on 4.7 and later

ADAPTIVE THINKING

Claude decides per request

- type adaptive, or on by default
- Thinks only where it helps
- Steered by the effort level
- Interleaves with tool calls automatically

THE TIEBREAKER

On current models, steer thinking with effort, never with a token budget.

Asking for adaptive thinking and setting effort

```python
response = client.messages.create(
    model="claude-opus-4-8",
    max_tokens=16000,
    thinking={"type": "adaptive"},
    output_config={"effort": "medium"},
    messages=[{"role": "user", "content": question}],
)
```

3 Leave room: thinking counts against max_tokens.

4 Needed on Opus 4.8; the 5.x models think by default.

5 Effort lives in output_config, not inside thinking.

ON THE EXAM

Anthropic: lower the effort level before reaching for prompt-based steering.

What thinking costs, and how to carry it

- **Billed as output tokens,** for the full reasoning, not just what you see.
- **Display omitted or summarised** changes what you see, not what you pay. Omitted cuts latency, not cost.
- **Earlier thinking kept in the history** is billed again as input.
- **In tool loops,** pass every thinking block back complete and unmodified. Edited blocks return a 400 error.
- **Changing thinking or effort** invalidates the prompt cache.

> ON THE EXAM
>
> Thinking is real output. Budget for it, carry it intact, and keep settings stable to keep the cache.

Effort: the main dial

LEVEL	USE IT FOR
`low`	Simple, high-volume work: classification, extraction
`medium`	Everyday tasks; the default on Opus 5.5 and Haiku 5.5
`high`	Harder reasoning; the default on Sonnet 5.5 and Fable 5.1
`xhigh`	Difficult, multi-step problems
`max`	The hardest work, where cost and time matter least

ON THE EXAM

Effort is "a behavioral signal, not a strict token budget". It shapes every token: thinking, tools and text.

Using effort well

- **Run an effort sweep** on your own evaluation set, and pick the lowest level that holds quality.
- **Shallow reasoning?** Raise effort rather than prompting around it.
- **Too low in agent loops:** at low effort Haiku 5.5 is more likely to skip a search or stop early.
- **Try effort before changing model.** It is the cheaper experiment.

> THE TIEBREAKER
>
> Effort is tuned with evidence, like everything else: sweep it on your test set.

Fast mode

- **Generally available on Opus 5.5** since 15 October 2026: set speed to fast, no beta header.
- **Up to 2.5x more output tokens per second,** with no change to intelligence.
- **Twice the price:** USD 8 in and 40 out per million tokens.
- **Faster output, not a faster first token.** Not with Batch; not on Claude Platform on AWS.
- **Switching speed** invalidates the prompt cache.

THE TAKEAWAY

Fast mode shortens long generations for people who are waiting, at a price.

Which dial for which problem?

THE PROBLEM	THE DIAL
Format or tone keeps varying	Examples, three to five, diverse
Known edge cases fail	An example of each edge case
Reasoning is shallow	Raise effort
Simple, high-volume work is too slow or costly	Lower effort, or a smaller tier
Long answers take too long on Opus	Fast mode, if the budget allows
The first words take too long	Streaming, caching, a smaller model

THE TIEBREAKER

Each dial fixes one kind of problem. Name the problem first.

> A support team has 40 hand-labelled examples showing exactly how case summaries should look. Claude's summaries are accurate but vary in layout from one case to the next.
>
> The team wants a consistent layout this week.

What should the developer do?

A Train a custom model on the forty labelled summaries before going further

B Store the examples in a database so staff can compare outputs against them

C Add an instruction asking Claude to be consistent in its layout

D Put three to five diverse examples in the prompt, each wrapped in example tags

◆ PAUSE AND DECIDE

What should the developer do?

A Train a custom model on the forty labelled summaries before going further

Heavy, slow, and unnecessary when the examples can go in the prompt.

B Store the examples in a database so staff can compare outputs against them

Claude never sees examples parked in a database.

C Add an instruction asking Claude to be consistent in its layout

Consistent with what? The layout is still undescribed.

D Put three to five diverse examples in the prompt, each wrapped in example tags

Multi-shot is the tool for consistent format. This week, no training.

THE SIGNAL

Labelled examples plus inconsistent format is the multi-shot signal.

> After four examples were added to Copperlet's tenant-reply prompt, the format improved. But now nearly every reply is three sentences long and opens with "Thanks for letting us know", even for urgent safety questions.
>
> All four examples happen to be three sentences long and open that way.

What is the best fix?

A Vary the examples in length, opening and situation, including an urgent safety case

B Remove the instructions and rely on the examples alone

C Add six more examples written in the same style as the first four

D Raise the effort level so Claude thinks harder about each reply

◆ PAUSE AND DECIDE

What is the best fix?

A Vary the examples in length, opening and situation, including an urgent safety case

Removes the accidental pattern and shows the edge case directly.

B Remove the instructions and rely on the examples alone

Loses the rules and keeps the pattern that caused the problem.

C Add six more examples written in the same style as the first four

Strengthens the very pattern being copied.

D Raise the effort level so Claude thinks harder about each reply

Effort fixes shallow reasoning, not imitation of examples.

THE SIGNAL

Outputs that mirror the examples' surface means the examples are not diverse enough.

> A team is writing an internal guide on thinking for developers who use current Claude models through the API.
>
> A reviewer asks them to keep only statements that are accurate today.

Which two statements are accurate? Choose two.

A. A thinking token budget is the recommended way to control thinking on Claude four point seven and later

B. Thinking is billed as output tokens even when the visible thinking text is omitted

C. Setting the thinking display to omitted makes each request cheaper

D. Thinking blocks can be shortened before being passed back in a tool loop, to save tokens

E. On Opus five point five thinking is always on, so effort is the lever for speed and cost

◆ PAUSE AND DECIDE

Which two statements are accurate? Choose two.

A. A thinking token budget is the recommended way to control thinking on Claude four point seven and later

Those models reject the budget with a 400. Use effort.

B. Thinking is billed as output tokens even when the visible thinking text is omitted

You pay for the full reasoning, whatever the display.

C. Setting the thinking display to omitted makes each request cheaper

It cuts latency, not cost.

D. Thinking blocks can be shortened before being passed back in a tool loop, to save tokens

Modified thinking blocks are rejected with a 400.

E. On Opus five point five thinking is always on, so effort is the lever for speed and cost

No off switch on that model; effort is the dial.

THE SIGNAL

Billing follows the reasoning, not the display. Control follows effort, not budgets.

> Copperlet's agency-ops agent runs on Opus 5.5 with tools. To save tokens, a developer's loop deletes the text inside each thinking block before sending the assistant turn back with the tool results.
>
> Every second request now fails with a 400 error.

What should the developer do?

A Turn thinking off for the agent so there are no thinking blocks to handle

B Set a lower temperature so the thinking blocks come back shorter

C Send back only the text blocks from the assistant turn, without any thinking blocks

D Pass the assistant turn back unchanged, thinking blocks included, alongside the tool results

◆ PAUSE AND DECIDE

What should the developer do?

A Turn thinking off for the agent so there are no thinking blocks to handle

Impossible on Opus 5.5: thinking is always on.

B Set a lower temperature so the thinking blocks come back shorter

Rejected on current models, and unrelated to the error.

C Send back only the text blocks from the assistant turn, without any thinking blocks

Still alters the turn Claude produced. Thinking travels with tool use.

D Pass the assistant turn back unchanged, thinking blocks included, alongside the tool results

Anthropic: pass every thinking block back complete and unmodified.

THE SIGNAL

In tool loops, the assistant's turn goes back exactly as it arrived. Save tokens with effort instead.

> Copperlet's triage feature classifies about 9,000 repair requests a week on Haiku 5.5 at its default effort. Accuracy on the 200-ticket test set is well above the target.
>
> The team wants it cheaper and faster without losing accuracy.

What should the team try first?

A Remove the examples from the prompt to save input tokens

B Move the feature to Opus so it can finish each ticket in fewer steps

C Run an effort sweep on the test set, and use low effort if accuracy still meets the target

D Turn on fast mode for every triage request

◆ PAUSE AND DECIDE

What should the team try first?

A Remove the examples from the prompt to save input tokens

Risks the accuracy the examples bought, for a small saving.

B Move the feature to Opus so it can finish each ticket in fewer steps

Many times the price per token, for headroom nobody needs.

C Run an effort sweep on the test set, and use low effort if accuracy still meets the target

The cheapest, measured experiment. Low effort suits simple classification.

D Turn on fast mode for every triage request

Opus 5.5 only, at twice the price. Wrong model, wrong goal.

THE SIGNAL

Accuracy to spare on simple, high-volume work: sweep effort downward, with the test set as judge.

> An internal tool streams long investment reports from Opus 5.5 to analysts. Text starts appearing quickly, but a full report takes over a minute, and analysts wait for the end before acting.
>
> The budget allows paying more for speed. The tool calls the Claude API directly.

What should the team do?

A Send each report request through the Message Batches API

B Turn on fast mode for this feature

C Lower max tokens so that reports come back shorter

D Add an instruction asking Claude to respond more quickly

◆ PAUSE AND DECIDE

What should the team do?

A Send each report request through the Message Batches API

Batch is for nobody waiting, and it is slower, not faster.

B Turn on fast mode for this feature

Up to 2.5x faster output on Opus 5.5, at a price the budget allows.

C Lower max tokens so that reports come back shorter

Cuts reports off. The length is the requirement.

D Add an instruction asking Claude to respond more quickly

Words in a prompt do not change generation speed.

THE SIGNAL

First words fast, total time slow, budget available, Opus on the Claude API: fast mode.

Module 6 in five lines

- **Examples fix format and edge cases:** three to five, relevant, diverse, in tags.
- **Thinking is billed output.** On the newest models it is on by default; Opus 5.5 cannot turn it off.
- **Adaptive thinking plus effort** replaces token budgets, which 4.7 and later reject.
- **Effort is the main dial.** Sweep it on your test set; try it before changing model.
- **Fast mode** speeds Opus 5.5 output at twice the price - not the first token.

COMING UP

Next: output handling - structured outputs, validation and defensive parsing.